

Date	

AIM:

ALGORITHM:

CODE:

```
import java.util.*;
import java.security.*;
class Main {

    static long power(long a, long b, long m) {
        long r = 1;
        for (int i = 0; i < b; i++)
```

```
        r = (r * a) % m;
    return r;
}
```

```
static long inverse(long a, long m) {
    for (long i = 1; i < m; i++)
        if ((a * i) % m == 1)
            return i;
    return 0;
}
```

```
static long hash(String msg, long q) throws Exception {
    MessageDigest md = MessageDigest.getInstance("SHA-256");
    byte[] h = md.digest(msg.getBytes());

    long value = 0;
    for (byte b : h)
        value = (value * 256 + (b & 255)) % q;

    return value;
}
```

```
public static void main(String[] args) throws Exception {

    Scanner sc = new Scanner(System.in);

    long p = 23, q = 11, g = 4;
```

```
long privateKey = 3;
```

```
long publicKey = power(g, privateKey, p);
```

```
long k = 7;
```

```
System.out.print("Enter the message: ");
```

```
String message = sc.nextLine();
```

```
long h = hash(message, q);
```

```
// Signature Generation
```

```
long r = power(g, k, p) % q;
```

```
long s = (inverse(k, q) * (h + privateKey * r)) % q;
```

```
System.out.println("\n--- DSA Parameters ---");
```

```
System.out.println("p = " + p);
```

```
System.out.println("q = " + q);
```

```
System.out.println("g = " + g);
```

```
System.out.println("\n--- Keys ---");
```

```
System.out.println("Private Key = " + privateKey);
```

```
System.out.println("Public Key = " + publicKey);
```

```
System.out.println("\n--- Signature ---");
```

```
System.out.println("Hash = " + h);
```

```
System.out.println("r = " + r);
```

```
System.out.println("s = " + s);
```

```
System.out.println("Signature = (" + r + "," + s + ")");
```

```

long w = inverse(s, q);

long u1 = (h * w) % q;

long u2 = (r * w) % q;


long v = (power(g, u1, p) * power(publicKey, u2, p) % p) % q;


System.out.println("\n--- Verification ---");

System.out.println("u1 = " + u1);

System.out.println("u2 = " + u2);

System.out.println("v = " + v);


if (v == r)

    System.out.println("Signature Status: VALID");

else

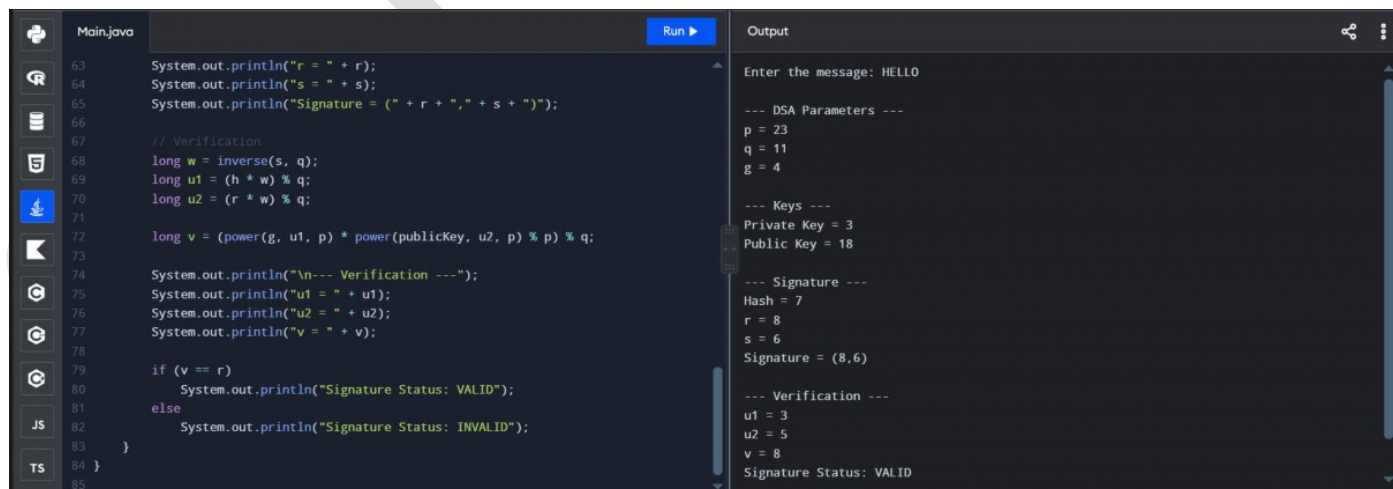
    System.out.println("Signature Status: INVALID");

}

}

```

Output:



The screenshot shows an IDE with a Java file named 'Main.java' and an 'Output' window. The code in the editor implements a DSA signature verification process. The output window shows the following text:

```

Enter the message: HELLO

--- DSA Parameters ---
p = 23
q = 11
g = 4

--- Keys ---
Private Key = 3
Public Key = 18

--- Signature ---
Hash = 7
r = 8
s = 6
Signature = (8,6)

--- Verification ---
u1 = 3
u2 = 5
v = 8
Signature Status: VALID

```

Result: